MILITARY SPECIFICATION

CIRCUIT, DEMOLITION, FOR DEMOLITION KIT, CRATERING: M180, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts assembly, and packing for one type of Demolition Circuit for the M180 Demolition Cratering Kit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-650 - Explosive: Sampling, Inspection and Testing.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

9244404- Circuit, Demolition.
9294600 - Packing and Marking for Box, Wood, Packing, Ammunition, for Demolition Circuit Sub-Pack M180.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with the applicable dawings, specifications and standards.

3.2 Components and assembly. The components and assembly shall comply with all requirements specified on drawing (dwg.) 9244404, and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Functioning. The demolition circuit shall function when conditioned and functioned in accordance with Para. 4.5.2.

3.4 Non-functioning. The assembly shall not function when subjected to test specified in Para. 4.5.1.

3.5 Pull test. There shall be no slippage between the detonating cord and the relay cup loading assembly when subjected to a minimum static load of twenty (20) pounds.

3.6 Radiographic examination. The gap between the detonating cord and the PETN charge of the relay cup loading assembly shall not exceed .03 inches when subjected to radiographic examination.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products

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and other foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample (see 6.2b) as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 25 complete sets of parts and sub-assemblies (i.e. 25 each of every component and sub-assembly, including at least five (5) items from each cavity used to produce molded parts and be so identified as to permit tracing each item to the cavity in which it was molded) and 50 loaded demolition circuit assemblies.

4.3.2 Inspections to be performed. See MIL-A-48078.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Circuit, Demolition, M180 Cratering Demolition Kit				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Relay Cup, Loading Assembly (Dwg. 8847494)				
	Examination for Defects	25		3.2	4.4.2.1
	Demolition Circuit Assembly (Dwg. 9244404)				
	Examination for Defects	50		3.2	4.4.2.2
	Non-function Test	50		3.4	4.5.1
	Function Test				
	Low Temperature	12		3.3	4.5.2.1
	Ambient Temperature	26		3.3	4.5.2.2
	High Temperature	12		3.3	4.5.2.3
	Pull Test	10		3.5	4.5.3
	Radiographic Examination	50		3.6	4.5.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Demolition, Circuits shall contain:

a. Electric Blasting Caps and connector assemblies from one lot interfix number of one manufacturer.

b. M-7 Non-Electric Blasting Caps from one lot interfix number of one manufacturer.

c. Relay Cup Loading Assemblies from one lot interfix number of one manufacturer.

d. Parts from not more than one lot interfix number from one manufacturer.

4.4.2. Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans with major and minor defects shall be in accordance with MIL-STD-105, inspection level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8847494
4.4.2.1	Relay Cup, Loading Assembly				NEXT HIGHER ASSEMBLY 9244404
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height of explosive incorrect		0.40%	3.2	Gage
102	Moisture content		Note 1	3.2	4.5.5
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual
	Notes: 1. Sampling shall be as specified in 4.5.5.				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Demolition Circuit Assembly		9244404
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shorting plug missing or loose		100%	3.2	Visual/Manual
2	Non-functioning test		100%	3.4	4.5.1
Major					
101	Length of detonating cord		0.40%	3.2	Gage
102	Priming adapter improperly assembled to assembly		0.40%	3.2	Visual
103	Detonating cord improperly assembled to electric blasting cap		0.40%	3.2	Visual/Manual
104	Insulation of wire leads damaged		0.40%	3.2	Visual
105	Crimps on non-electric blasting cap less than 360°		0.40%	3.2	Visual
106	Crimps on relay cup loading assembly less than 360°		0.40%	3.2	Visual
107	Relay cup loading assembly pull test		100%	3.5	4.5.3
108	Radiographic examination		0.40%	3.6	4.5.4
109	Component missing or damaged		0.40%	3.2	Visual
110	Functioning			3.3	4.5.2
111	Low temperature	20(a)		3.3	4.5.2.1
112	Ambient temperature	40(a)		3.3	4.5.2.2
113	High temperature	20(a)		3.3	4.5.2.3

NOTES:
a. If the amount of rejects totals three (3) or more samples, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Demolition Circuit Assembly (cont'd)				9244404
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Color of wire		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Wooden Box, Prior to Sealing				9294600
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly not dry prior to packaging		0.40%	3.2	Visual
102	Plastic bag missing or damaged or improperly sealed (dwg. 9289289)		0.40%	3.2	Visual
103	Board broken or with split terminating at edge		0.40%	3.2	Visual
104	Assembly missing		0.40%	3.2	Visual
105	Contents not tight		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Sealed Wooden Box		9294600
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing, broken, or loose		0.40%	3.2	Visual
102	Board broken or with split terminating at edge		0.40%	3.2	Visual
103	Contents of box exposed		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual
203	Strapping covers marking		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48853B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Packing and Marking for Storage and Shipments (when applicable)		9294600
			NEXT HIGHER ASSEMBLY 8808362

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Nomenclature missing, illegible or incorrect		100%	3.2	Visual
102	DOT markings missing, illegible or incorrect		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection Equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraph (See 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Non-functioning. The demolition circuits shall be tested by utilizing pulse producing equipment to apply a pulse having energy of 3.0 milliwatt seconds, plus or minus 1.0 milliwatt second for not less than 10.0 nor more than 24.0 milliseconds. Any assembly which fails to comply with the requirement specified in 3.5 shall be classed defective and removed from the lot. Test equipment shall be as specified in 4.4.4 (non-destructive test).

4.5.2 Functioning.

4.5.2.1 Low temperature (-40°F). The demolition circuits shall be conditioned at a temperature of minus 40 degrees, plus 5 degrees F for a minimum of 24 hours. They shall be functioned within 5 minutes after removal from the temperature chamber, utilizing pulse producing equipment to apply a pulse of 18.0 plus or minus 1.0 milliwatts for not less than 10.0 or more than 24.0 milliseconds. Any assembly which fails to comply with the requirements specified in 3.3, shall be classed defective.

4.5.2.2 Ambient temperature (+70°F). The demolition circuits shall be conditioned at a temperature of 70 degrees plus or minus 5 degrees F for a minimum of 6 hours. They shall be functioned within 5 minutes after removal from the temperature chamber utilizing pulse producing equipment to apply a pulse of 18.0 plus or minus 1.0 milliwatt seconds for not less than 10.0 or more than 24.0 milliseconds. Any assembly which fails to comply with the requirements specified in 3.3, shall be classed defective. Test equipment shall be as specified in 4.4.4.

4.5.2.3 High temperature (+130°F). The demolition circuits shall be conditioned at a temperature of +130 degrees plus 5 degrees F for a minimum of 24 hours. They shall be functioned within 5 minutes after removal from the temperature chamber utilizing pulse producing equipment to apply a pulse of 18.0 plus or minus 1.0 milliwatt seconds for not less than 10.0 or more than 24.0 milliseconds. Any assembly which fails to comply with the requirements specified in 3.3, shall be classed defective. Test equipment shall be as specified in 4.4.4.

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4.5.3 Relay cup loading assembly pull test. The relay cup loading assembly and the detonation cord shall be held securely and the required load applied in a smooth, gradual manner. Any assembly which fails to comply with the requirement specified in 3.5 shall be classed defective and removed from the lot. Test equipment shall be as specified in 4.4.4.

4.5.4 Radiographic examination. The assembly shall undergo radiographic examination using approved test equipment as specified in 4.4.4. Any assembly which fails to comply with the requirements specified in 3.6, shall be classed defective.

4.5.5 Moisture content of explosive charge. The moisture content of the explosive charge shall be determined as specified in MIL-STD-650, Method 101.6. The sample shall be selected from that quantity of charge to be used to produce a lot of loaded relay cups.

5. PACKAGING

5.1 Packaging (level B). Packaging of the demolition circuits shall be in accordance with Dwg. 8808362.

5.2 Packing (level B). Packing shall in accordance with Dwgs. 8808362 and 9294600.

5.3 Marking. Marking shall be in accordance with Dwgs. 8808362 and 9294600.

5.4 Shipping. When circuits from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. This demolition circuit is used as part of the M180 Cratering Demolition Kit.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAT, Dover, New Jersey 07801.

Custodian:
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Preparing Activity:
Army-AR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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